

1 Setup B

Optimal thresholds. At $\alpha = 1/2$, maximizing J_{EE}^N over δ is minimizing

$$M_S(\delta) = (q_- - q_+) + 2\sqrt{(1 - q_-)q_+}, \quad q_{\pm} = Q\left(\frac{\mu/2 \pm \delta}{\sigma}\right). \quad (1)$$

Write $\phi_{\pm} \triangleq \phi\left(\frac{\mu/2 \pm \delta}{\sigma}\right)$, $A \triangleq \sqrt{q_+}$, $B \triangleq \sqrt{1 - q_-}$.

$$\sigma \frac{dM_S}{d\delta} = (\phi_- + \phi_+) - \frac{\phi_- A^2 + \phi_+ B^2}{AB} = \frac{(B - A)(\phi_- A - \phi_+ B)}{AB}. \quad (2)$$

$B > A$ always: $B^2 = Q\left(\frac{\delta - \mu/2}{\sigma}\right) > Q\left(\frac{\delta + \mu/2}{\sigma}\right) = A^2$. So the sign is carried entirely by $\phi_- A - \phi_+ B$, and stationarity is

$$\frac{\phi_-}{\phi_+} = \frac{B}{A} \iff e^{\mu\delta/\sigma^2} = \sqrt{\frac{1 - q_-}{q_+}}. \quad (3)$$

Since $\delta = \frac{\sigma^2}{\mu} \log \tau_2$, the left side is τ_2 , so the optimal thresholds satisfy

$$\boxed{\tau_2^{*2} = \frac{P\gamma^S(u^1|H_2)}{P\gamma^S(u^2|H_1)}} \quad (4)$$

The full derivation, are given in Appendix 2.1.

2 Appendix

2.1 Optimal Silence Threshold

The objective. Substituting $q_{\pm} = Q\left(\frac{\mu/2 \pm \delta}{\sigma}\right)$ into (1):

$$M_S(\delta) = \left[Q\left(\frac{\mu/2 - \delta}{\sigma}\right) - Q\left(\frac{\mu/2 + \delta}{\sigma}\right)\right] + 2\sqrt{\left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right] Q\left(\frac{\mu/2 + \delta}{\sigma}\right)}. \quad (5)$$

Using $1 - Q(x) = Q(-x)$, equivalently

$$M_S(\delta) = 1 - Q\left(\frac{\delta - \mu/2}{\sigma}\right) - Q\left(\frac{\delta + \mu/2}{\sigma}\right) + 2\sqrt{Q\left(\frac{\delta - \mu/2}{\sigma}\right) Q\left(\frac{\delta + \mu/2}{\sigma}\right)}, \quad (6)$$

giving $M_S(0) = 2\sqrt{q(1 - q)} = M_V$ and $M_S(\delta) \rightarrow 1$ as $\delta \rightarrow \infty$.

Differentiation. $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$, so $Q'(x) = -\phi(x)$ with $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. The inner derivatives are $\mp 1/\sigma$, so

$$\frac{d}{d\delta} \left[Q\left(\frac{\mu/2 - \delta}{\sigma}\right) - Q\left(\frac{\mu/2 + \delta}{\sigma}\right)\right] = \frac{1}{\sigma} \left[\phi\left(\frac{\mu/2 - \delta}{\sigma}\right) + \phi\left(\frac{\mu/2 + \delta}{\sigma}\right)\right]. \quad (7)$$

For $f(\delta) \triangleq \left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right] Q\left(\frac{\mu/2 + \delta}{\sigma}\right)$,

$$f'(\delta) = -\frac{1}{\sigma} \phi\left(\frac{\mu/2 - \delta}{\sigma}\right) Q\left(\frac{\mu/2 + \delta}{\sigma}\right) - \frac{1}{\sigma} \phi\left(\frac{\mu/2 + \delta}{\sigma}\right) \left[1 - Q\left(\frac{\mu/2 - \delta}{\sigma}\right)\right], \quad (8)$$

and $\frac{d}{d\delta}2\sqrt{f} = f'/\sqrt{f}$. Combining:

$$\sigma \frac{dM_S}{d\delta} = \phi\left(\frac{\mu/2-\delta}{\sigma}\right) + \phi\left(\frac{\mu/2+\delta}{\sigma}\right) - \frac{\phi\left(\frac{\mu/2-\delta}{\sigma}\right) Q\left(\frac{\mu/2+\delta}{\sigma}\right) + \phi\left(\frac{\mu/2+\delta}{\sigma}\right) \left[1 - Q\left(\frac{\mu/2-\delta}{\sigma}\right)\right]}{\sqrt{\left[1 - Q\left(\frac{\mu/2-\delta}{\sigma}\right)\right] Q\left(\frac{\mu/2+\delta}{\sigma}\right)}}, \quad (9)$$

which is (2) written out.

Factorization. Multiplying by $AB > 0$:

$$\begin{aligned} (\phi_- + \phi_+)AB - (\phi_-A^2 + \phi_+B^2) &= \phi_-A(B - A) + \phi_+B(A - B) \\ &= (B - A)(\phi_-A - \phi_+B). \end{aligned} \quad (10)$$

Since $B - A > 0$, setting $M'_S = 0$ forces $\phi_-A = \phi_+B$. The density ratio is closed form,

$$\frac{\phi_-}{\phi_+} = \exp\left(\frac{(\mu/2 + \delta)^2 - (\mu/2 - \delta)^2}{2\sigma^2}\right) = e^{\mu\delta/\sigma^2}, \quad (11)$$

which gives (3).